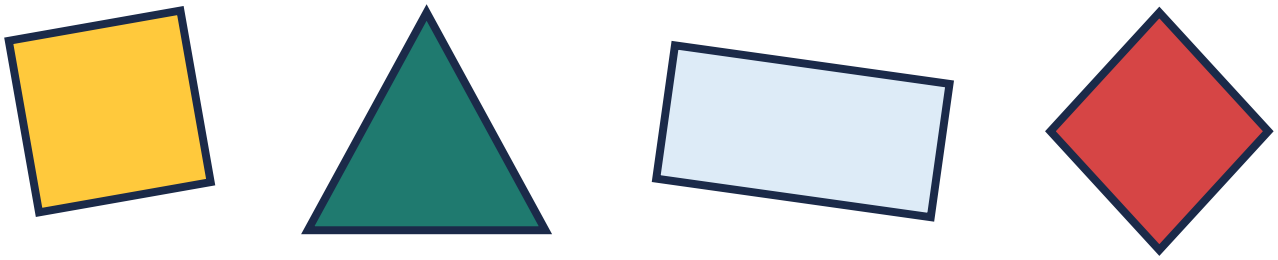
FOR PARENTS • PRIMARY 3 HOLIDAY PROGRAM

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

The three levels

Core is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Number Sleuths

LESSON 1 • NUMBERS TO 10 000

Core

1 Write the number. Think of 6 big boxes of 1000 sweets, 4 packets of 10 sweets and 9 loose sweets. (There are no boxes of 100.)

2 Look at the place value chart: Th 2 | H 8 | T 3 | O 7. This is the number 2837. How much is the digit 8 worth?

3 Use the cards 9, 1, 4, 6 once each. Make the smallest 4-digit number.

4 What number is 10 less than 5003?

Think Harder

5 Use the cards 5, 0, 8, 2 once each. Make the greatest even number and the smallest odd number.

6 How many 4-digit numbers can you make with the cards 1, 1, 0, 0? List them.

What your child learns

4-digit numbers: place value, the greatest and smallest numbers from digit cards, listing every number in order, and number patterns.

Tip for you

Write four digits on scraps of paper. Ask for the greatest odd number, then the number closest to 5000.

Ask

"How do you know you have found them all?"

7 A number has 7 thousands, 13 hundreds and 6 ones. What is the number?

Challenge

8 Ali writes a 4-digit number using only odd digits, all different. It is as close to 5000 as possible. What is it?

9 A pattern starts at 1000 and goes up by 7 each time: 1000, 1007, 1014, ... Is 1100 in the pattern? Explain.

W1 Cryptarithm Crackers

LESSON 2 • ADDITION AND SUBTRACTION

Core

1 A library has 3999 storybooks and 2456 comic books. How many books are there altogether? Write the numbers in a Th H T O grid and add the ones first.

2 A shop has 7000 stickers and sells 2468 of them. How many are left? Hint: draw a number line from 2468 to 7000 and jump to 2500, then to 3000, then to 7000. Add up the jumps.

3 Find the missing digit. Add in columns, starting with the ones, or try the digits one at a time. 4□27 + 1365 = 5992

4 Find the missing digit. Try 5, 6 and 7 and check each one by taking away. 8□50 – 3420 = 5230

Think Harder

5 Find the missing digits: 2□7□ + □6□9 = 6000

6 In a number pyramid, each brick is the sum of the two bricks under it. The bottom row is 1200, □, □. The middle row is 2650, □. The top brick is 6000. Find the missing bricks.

What your child learns

Missing digits in 4-digit sums, checking subtraction by adding, magic squares and number pyramids.

Tip for you

Cover one digit of a long sum with your thumb. Can your child find it and check by adding back?

Ask

"How can you check your answer?"

7 Ali meant to add 1099 to a number, but he added 1999 by mistake and got 6543. What is the correct answer?

Challenge

8 Find the missing digits: 8□0□ – □4□7 = 2468

9 A 4-digit number has all different digits that add up to 30. Find the difference between the greatest and the smallest such number.

W2 LESSON 3 • TABLES OF 6, 7, 8 AND 9
Times Table Treasure

Core

1 Draw 6 rows of 8 dots. How many dots are there? Now turn your drawing sideways: it shows 8 rows of 6 dots. How many dots now? What do you notice?

2 You know 5 × 9 = 45. Use it to find 6 × 9.

3 63 apples are packed into boxes of 7 apples with none left over. Count in 7s (7, 14, 21, ...) to find how many boxes there are.

4 A spider has 8 legs. How many legs do 7 spiders have?

Think Harder

5 The digits of every answer in the 9 times table (9 × 1 to 9 × 10) add up to 9. Which of these are in the 9 times table: 54, 65, 72, 83?

6 What is the last digit of 8 × 8 × 8? Use last digits only.

What your child learns

Patterns inside the 6, 7, 8 and 9 times tables, last-digit patterns, and splitting arrays into facts they know.

Tip for you

Ask for 7 × 8 split into two easier facts, such as 5 × 8 and 2 × 8.

Ask

"What pattern do you notice in the last digits?"

7 Ravi says: "Every answer in the 6 times table is even." Is he right? Explain why.

Challenge

8 Four numbers in a multiplication square are 42 and 48 in the top row, and 49 and 56 in the row below. What are the two row numbers and the two column numbers?

9 Ben writes the answers of the 6 times table from 6 × 1 to 6 × 20. How many of the answers end in 8?

W2 Model Makers

LESSON 4 • WORD PROBLEMS: BAR MODELS

Core

1 There are 2345 people at a concert. 1180 of them are adults. How many are children? Draw one long bar for all the people and mark off the part for the adults.

2 A red ribbon is 145 cm long. A blue ribbon is 68 cm longer. Draw the red ribbon as a bar, then draw a longer bar for the blue ribbon. How long is the blue ribbon?

3 Ravi has 512 stamps. Siti has 139 fewer stamps than Ravi. Draw a bar for each child. How many stamps does Siti have? How many do they have altogether?

4 Ali has 23 cards. Ben has 4 times as many cards as Ali. Draw 1 box for Ali and 4 boxes for Ben. How many cards does Ben have? How many do they have altogether?

Think Harder

5 Ali and Ben have 96 cards altogether. Ali has 3 times as many cards as Ben. How many more cards does Ali have than Ben?

What your child learns

Drawing bar models for part-whole, comparison and before-and-after word problems.

Tip for you

When your child reads a word problem, ask them to draw the bars before doing any sums.

Ask

"What stays the same? What changes?"

6 Siti has 54 more beads than Mei. Siti gives Mei some beads, and now they have the same number. How many beads did Siti give Mei?

7 Three boxes hold 600 pens altogether. Box A has 50 more pens than Box B. Box C has 50 more pens than Box A. How many pens are in Box C?

Challenge

8 Ali and Ben have 120 stickers altogether. Ali has 3 times as many as Ben. Ali gives some stickers to Ben, and then Ali has only twice as many as Ben. How many stickers does Ali give to Ben?

9 A pencil box and 3 pens cost $25 altogether. A pencil box and 5 pens cost $35 altogether. How much does one pen cost? How much does the pencil box cost?

W3 LESSON 5 • DIVISION
Remainder Riddles

Core

1 Share 35 counters equally among 4 children. How many counters does each child get? How many counters are left over? (Use counters or draw dots in 4 groups.)

2 Put 58 sweets into bags with 7 sweets in each bag. How many full bags can you fill? How many sweets are left over? (Use counters or draw dots.)

3 6 friends share 45 stickers equally. How many does each friend get? How many are left over?

4 30 children go on an outing. Each car holds 4 children. How many cars are needed?

Think Harder

5 Mei has 53 beads. She makes bracelets with 8 beads each. How many bracelets can she make? How many more beads does she need to make one more bracelet?

6 I am a number between 20 and 30. When I am divided by 4, the remainder is 3. When I am divided by 6, the remainder is 5. What number am I?

What your child learns

Division with remainders, and deciding what the leftover means in a story (round up or round down).

Tip for you

Ask: 30 people, 4 to a car. How many cars? Why is it not 7?

Ask

"What do you do with the leftover here?"

7 What is the biggest 2-digit number that leaves a remainder of 8 when it is divided by 9?

Challenge

8 I am a 2-digit number. When I am divided by 3, by 4 or by 5, the remainder is always 2. What number am I?

9 A box has more than 30 but fewer than 50 chocolates. They can be shared equally among 2, 3, 4 or 6 children with none left over. Find all the numbers of chocolates there could be.

W3 Fraction Walls

LESSON 6 • FRACTIONS

Core

1 A chocolate bar has 10 equal pieces. Ali eats half of the bar. How many pieces is that? Write it as 1/2 = ?/10. (Draw the bar and cut it in two halves if you like.)

2 Two pizzas are the same size. One is cut into 6 equal slices and the other into 9 equal slices. Which pizza has the bigger slices? So which is bigger, 1/6 of a pizza or 1/9 of a pizza?

3 A bar is cut into 9 equal parts. Shade 2 parts. Then shade 5 more parts. What fraction of the bar is shaded now?

4 A bar is cut into 8 equal parts and 3 parts are shaded. What fraction of the bar is NOT shaded? (Draw it and count.)

Think Harder

5 Which is bigger, 3/5 or 7/10? Show how you know.

6 Siti reads 1/4 of a book on Monday and 1/2 of the book on Tuesday. What fraction of the book has she read? What fraction is left?

What your child learns

Equivalent fractions with a fraction wall, comparing fractions, and adding parts of the same whole.

Tip for you

Fold paper strips into halves, quarters and eighths. Which pieces are the same size?

Ask

"Are the parts the same size? How do you know?"

7 Put these in order, smallest first: 2/3, 5/6, 7/12.

Challenge

8 Which is bigger, 3/7 or 5/9? Use 1/2 to help you.

9 Which is bigger, 2/5 or 4/9? Hint: make the top numbers the same.

W4 Area Architects

LESSON 7 • AREA AND PERIMETER

Core

1 A rectangle is 9 cm long and 4 cm wide. Walk all the way round its edge: how far is that? (This distance is called the perimeter.) Draw it on squared paper if you like.

2 Cover a rectangle 7 cm long and 6 cm wide with squares that are 1 cm by 1 cm. There are 6 rows with 7 squares in each row. How many squares cover it? (The number of squares is the area, in cm².)

3 A square has 4 equal sides. The distance all the way round it is 32 cm. How long is each side?

4 A square floor is covered with 36 small squares (1 cm by 1 cm), with the same number of squares in each row and each column. How many squares are there along one side? How far is it all the way round the edge?

Think Harder

5 Find all the rectangles with whole-cm sides and an area of 24 cm². Which one has the shortest perimeter?

6 A rectangle is 12 cm long. Its perimeter is 30 cm. Find its area.

What your child learns

Area and perimeter: same perimeter but different area, figures made of rectangles, and listing every rectangle.

Tip for you

Use 12 square biscuits or tiles. How many different rectangles can your child make? Which has the longest edge?

Ask

"Does a bigger area always mean a bigger perimeter?"

7 A rectangle 8 cm by 2 cm and a square have the same perimeter. Which has the bigger area, and by how much?

Challenge

8 List every rectangle with whole-cm sides and a perimeter of 24 cm, with its area, in a table. What do you notice as the rectangle gets closer to a square? Use what you notice to predict the biggest area for a perimeter of 40 cm, then check.

9 A rectangle is made of 2 same-size squares side by side. Its perimeter is 36 cm. What is its area?

W4

LESSON 8 • ANGLES, PERPENDICULAR AND PARALLEL LINES

Angle and Line Detectives

Core

1 A right angle is a square corner, like the corner of a page. How many right angles does a rectangle have? Check with the corner of a piece of paper.

2 At 3 o'clock the minute hand points at 12 and the hour hand points at 3. Is the angle between the hands smaller than, bigger than, or the same as the corner of a page?

3 At 2 o'clock, is the angle between the clock hands smaller or bigger than the corner of a page?

4 Parallel lines are straight lines that stay the same distance apart and never meet, like the two rails of a train track. Name 2 capital letters that have a pair of parallel lines.

Think Harder

5 I have 4 sides and 4 right angles. My sides are not all the same length. What shape am I? How many pairs of parallel lines do I have?

What your child learns

Right angles, counting angles in an organised way, and spotting parallel and perpendicular lines.

Tip for you

Fold a corner of paper to make a right-angle checker and go on a hunt around the house.

Ask

"How did you make sure you counted every angle?"

6 A U shape is made of 5 squares: 3 squares in a row with 1 more square on top of each end square. How many right angles are inside it?

7 Look at a clock face. For each o'clock time from 1 o'clock to 11 o'clock, is the angle between the hands (the short way round) smaller than, the same as, or bigger than a right angle? Put each time in the right group.

Challenge

8 4 lines from one point make 6 angles. How many lines from one point are needed to make exactly 10 angles?

9 A staircase is made of columns of 1, 2 and 3 squares standing side by side (3 steps). It has 6 right angles inside it. How many right angles are inside a staircase with 4 steps? With 5 steps? With 10 steps? Find a rule and explain why it works.

W5

LESSON 9 • LENGTH, MASS, VOLUME, MONEY AND TIME

Measure and Money Masters

Core

1 A ribbon is 2 m 45 cm long. (1 m is 100 cm.) How many cm is the ribbon?

2 A bag of rice weighs 3060 g. (1 kg is 1000 g.) Write its mass in kg and g.

3 A pail holds 4 ℓ 500 ml. A jug holds 1 ℓ 750 ml. (1 ℓ is 1000 ml.) How much more does the pail hold? Hint: write both in ml first.

4 Ali buys a sandwich for $3.80 and a drink for $1.45. He pays with a $10 note. How much change does he get?

Think Harder

5 A 2 ℓ bottle of juice is poured into cups. Each cup holds 300 ml. How many cups can be filled? How much juice is left over?

6 A shop opens at 09:30 and closes at 21:00. How long is it open?

What your child learns

Changing to one unit (cm, g, ml), budgets with money, durations and the 24-hour clock.

Tip for you

Plan a family outing together: how long will each part take, and what time will you be home?

Ask

"Did you change everything to the same unit first?"

7 A box with 4 books weighs 1 kg 50 g. The empty box weighs 250 g. All the books weigh the same. How heavy is one book?

Challenge

8 A 5 m ribbon is cut into two pieces. One piece is 80 cm longer than the other. How long is each piece?

9 You have a 300 ml cup, a 500 ml cup (no marks on either) and a tap. How can you get exactly 100 ml in a cup?

W5 Challenge Day

LESSON 10 • MIXED PUZZLES: BAR GRAPHS AND LOGIC

Core

1 Mei is 5th from the front and 4th from the back of a queue. Draw circles for the children in the queue. How many children are in the queue?

2 Two numbers add up to 50. One number is 12 more than the other. Guess a pair, check it, and improve your guess. What are the two numbers?

3 Make all the 3-digit numbers you can with the digit cards 1, 4 and 7 (use each card once). Which of them are even?

4 A bar graph has lines at 0, 10, 20, 30, 40 and 50. A bar stops halfway between 30 and 40. What number does it show?

Think Harder

5 Ali has three coins: 10¢, 20¢ and 50¢. He picks one coin, or two coins, or all three. List every different amount of money he can make, in order. How many amounts are there?

What your child learns

Choosing how to start a puzzle: draw it, list it or make a table, plus bar graphs and logic grids.

Tip for you

When your child is stuck, ask "Can you draw it?" before giving any other hint.

Ask

"How did you start? What would you try next time?"

6 Ali, Ben and Siti each have a different pet: a cat, a dog and a fish. Ali's pet has no fur. Ben does not have the dog. Who has which pet?

7 Triangles are made in a row with sticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. How many sticks are needed for 12 triangles?

Challenge

8 Ben is 3rd from the front and 5th from the back of a queue, so there are 7 children. Add the two numbers: what do you get? Does this always work? Mei is 6th from the front in a queue of 15. Where is she from the back? Write a rule that links "from the front", "from the back" and "how many in the queue".

9 How many squares of any size are there in a 3 by 3 grid of squares?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Number Sleuths

1. 6049
2. 800
3. 1469
4. 4993
5. Greatest even: 8520. Smallest odd: 2085 (5 is the only odd card, so it goes last).
6. 3 numbers: 1001, 1010, 1100.
7. 8306 (7000 + 1300 + 6)
8. 5137 (137 more than 5000). The closest below 5000 is 3975, which is 1025 less.
9. No. 14 jumps of 7 is 98, so 1098 is in the pattern. The next number is 1105, so 1100 is skipped.

Week 1 • Lesson 2: Cryptarithm Crackers

1. 6455 (3999 + 2456; or add 4000 to get 6456, then take away 1)
2. 4532 (jumps of 32 + 500 + 4000 = 4532; or 7000 – 2468)
3. 6 (4627 + 1365 = 5992)
4. 6 (8650 – 3420 = 5230)
5. 2371 + 3629 = 6000 (the only way)
6. Bottom row 1200, 1450, 1900. Middle row 2650, 3350.
7. 5643 (he added 900 too much: 6543 – 900)
8. 8905 – 6437 = 2468 (the only way; check 2468 + 6437 = 8905)
9. 3087. The only four different digits that add up to 30 are 9, 8, 7, 6, so 9876 – 6789 = 3087.

Week 2 • Lesson 3: Times Table Treasure

1. Both are 48: 8 × 6 and 6 × 8 are the same. Swapping the numbers does not change the answer.
2. 54 (one more 9: 45 + 9).
3. 9 boxes (7 × 9 = 63)
4. 56
5. 54 and 72 (5 + 4 = 9 and 7 + 2 = 9; 6 + 5 and 8 + 3 make 11). 54 = 9 × 6, 72 = 9 × 8.
6. 2. 8 × 8 = 64 ends in 4, and 4 × 8 = 32 ends in 2. (8 × 8 × 8 = 512.)
7. Yes. 6 groups can always be put into 3 pairs of groups, so nothing is left over. (6 × 7 = 3 × 14 = 42.)
8. Rows 6 and 7, columns 7 and 8. Start with 49: it can only be 7 × 7.
9. 4 (18, 48, 78, 108). The last digits 6, 2, 8, 4, 0 repeat every 5, so 8 comes once in every 5.

Week 2 • Lesson 4: Model Makers

1. 1165
2. 213 cm
3. Siti 373; altogether 885
4. Ben 92; altogether 115
5. 48. 4 units = 96, 1 unit = 24, and the difference is 2 units.
6. 27 (half of the difference).
7. 250. Take away the extras (50 + 100 = 150): 450 ÷ 3 = 150 in Box B, so A has 200 and C has 250.
8. 10. At first 4 units = 120, so Ben 30 and Ali 90. The total is still 120, but now Ali has 2 parts and Ben 1 part, so 3 equal parts = 120 and each part is 40. Ben now has 40 and Ali 80. Ben went from 30 to 40, so Ali gave 10.
9. Pen $5, box $10. The second bill is $10 more because it has 2 more pens, so 2 pens cost $10 and 1 pen $5. Then 3 pens cost $15, so the box costs $25 – $15 = $10.

Week 3 • Lesson 5: Remainder Riddles

1. 8 each, 3 left over (4 × 8 = 32, 35 – 32 = 3)
2. 8 full bags, 2 sweets left over (7 × 8 = 56)
3. 7 each, 3 left over (6 × 7 = 42)
4. 8 cars (30 ÷ 4 = 7 r 2; the 2 children left over need a car too)
5. 6 bracelets (53 ÷ 8 = 6 r 5). She needs 3 more beads (56 – 53).
6. 23 (remainder 3 when ÷ 4: 23 and 27; 23 ÷ 6 = 3 r 5, 27 ÷ 6 = 4 r 3)
7. 98 (9 × 10 + 8 = 98; 99 ÷ 9 has no remainder)
8. 62 (60 is in the 3, 4 and 5 tables; 60 + 2 = 62; the next one, 122, has 3 digits)
9. 36 and 48 (numbers in the 12 times table)

Week 3 • Lesson 6: Fraction Walls

1. 5 (1/2 = 5/10)
2. 1/6 (cutting into fewer slices makes bigger slices)
3. 7/9 (2/9 + 5/9)
4. 5/8 (8/8 – 3/8)
5. 7/10 (3/5 = 6/10, and 7/10 is bigger than 6/10)
6. She has read 3/4 (1/4 + 2/4). 1/4 is left.
7. 7/12, 2/3, 5/6 (7/12, 8/12, 10/12)
8. 5/9. 3/7 is less than 1/2 (half of 7 is 3 and a half) and 5/9 is more than 1/2 (half of 9 is 4 and a half).
9. 4/9. 2/5 = 4/10, and ninths are bigger pieces than tenths, so 4/9 > 4/10.

Week 4 • Lesson 7: Area Architects

1. 26 cm (9 + 4 + 9 + 4)
2. 42 cm² (6 rows of 7)
3. 8 cm (32 ÷ 4)
4. 6 squares along one side (6 × 6 = 36), so the distance round is 6 + 6 + 6 + 6 = 24 cm.
5. 24 by 1 (50 cm), 12 by 2 (28 cm), 8 by 3 (22 cm), 6 by 4 (20 cm). The 6 cm by 4 cm rectangle has the shortest perimeter.
6. Length + breadth = 15, so the breadth is 3 cm. Area 12 × 3 = 36 cm².
7. Perimeter 20 cm, so the square has sides of 5 cm. Square 25 cm², rectangle 16 cm². The square, by 9 cm².
8. 11 by 1 (11 cm²), 10 by 2 (20), 9 by 3 (27), 8 by 4 (32), 7 by 5 (35), 6 by 6 (36). The area grows as the sides get closer to each other, and the square has the biggest area. For 40 cm the square is 10 by 10, area 100 cm² (the next best, 11 by 9, is 99 cm²).
9. The perimeter is 6 sides of a small square, so each side is 6 cm. Area 2 × 36 = 72 cm².

Week 4 • Lesson 8: Angle and Line Detectives

1. 4
2. The same: a right angle
3. Smaller
4. For example H and E (also F, N, Z, U, M).
5. A rectangle. 2 pairs of parallel lines.
6. 6 (it has 8 corners; the 2 corners at the bottom of the gap are bigger than a right angle inside the shape).
7. Smaller than a right angle: 1, 2, 10 and 11 o'clock. A right angle: 3 and 9 o'clock. Bigger: 4, 5, 6, 7 and 8 o'clock (at 6 o'clock the hands make a straight line).
8. 5 lines (4 + 3 + 2 + 1 = 10).
9. 4 steps: 7. 5 steps: 8. 10 steps: 13. The rule is: number of steps + 3. Each column has one square corner at its top right (that is the number of steps), the tallest column has one more at its top left, and there are 2 square corners at the bottom. The inside corners where one step meets the next are dents (bigger than a right angle), so they do not count.

Week 5 • Lesson 9: Measure and Money Masters

1. 245 cm
2. 3 kg 60 g
3. 2 ℓ 750 ml (4500 ml − 1750 ml = 2750 ml)
4. $4.75 ($3.80 + $1.45 = $5.25; $10 − $5.25 = $4.75)
5. 6 cups, 200 ml left (300, 600, 900, 1200, 1500, 1800 ml; 2000 − 1800 = 200 ml)
6. 11 h 30 min (30 min to 10:00, then 11 h to 21:00)
7. 200 g (1050 g − 250 g = 800 g; 800 ÷ 4 = 200 g)
8. 2 m 10 cm and 2 m 90 cm (500 − 80 = 420; 420 ÷ 2 = 210)
9. Fill the 300 ml cup and pour it into the 500 ml cup. Fill the 300 ml cup again and pour until the 500 ml cup is full (200 ml goes in). 100 ml is left in the 300 ml cup.

Week 5 • Lesson 10: Challenge Day

1. 8 (5 + 4 − 1: Mei is counted in both)
2. 19 and 31 (19 + 31 = 50 and 31 − 19 = 12; or 50 − 12 = 38 and 38 ÷ 2 = 19)
3. 147, 174, 417, 471, 714, 741. Even: 174 and 714.
4. 35
5. 7 amounts: 10¢, 20¢, 30¢ (10 + 20), 50¢, 60¢ (10 + 50), 70¢ (20 + 50), 80¢ (all three). Listing by the number of coins (one, two, three) makes sure none is missed.
6. Ali: fish. Ben: cat. Siti: dog.
7. 25 sticks (add 2 each time: 3 + 11 × 2 = 25)
8. 3 + 5 = 8, which is 1 more than 7. It always works: the child in question is counted in both numbers, so (from the front) + (from the back) = (children in the queue) + 1. Mei is 15 + 1 − 6 = 10th from the back.
9. 14 (9 small, 4 of size 2 by 2, 1 of size 3 by 3)